

Nima Kelidari

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Education

University of Southern California - Viterbi School of Engineering Los Angeles, CA
M.S. in Computer Science - Artificial Intelligence | GPA: 3.55/4.0 (3.8/4.0 in AI/ML) Dec 2024 – Dec 2026 (expected)
• Coursework: Machine Learning, Deep Learning, **Multimodal Probabilistic Learning**, Applied NLP, **Robot Manipulation**, Analysis of Algorithms
• **Technical Skills – Languages:** Python (Proficient), Java (Proficient), C++, JavaScript, SQL | **ML:** PyTorch, JAX, TensorFlow, Hugging Face, vLLM, OpenCV, CUDA/CuPy, MuJoCo, ROS | **Infrastructure:** SLURM, Docker, Kubernetes, Jenkins, MLflow, Weights & Biases, DVC, FastAPI, GCP, AWS, Git | **Concepts:** ML & NLP, numerical optimization, RL & imitation learning, **distributed systems**, information retrieval, computer vision

Sharif University of Technology Tehran, TE
Bachelor of Science in Computer Science | GPA: 3.37/4.0 Sep 2019 – Feb 2024
• 3.5/4.0 across 15+ CS/AI courses: Data Structures, Algorithm Analysis, Image Processing, Computer Vision, Deep Learning, Regression Analysis

Experience

Memorial Sloan Kettering Cancer Center - Department of Medical Physics New York, NY (Remote)
Computational Optimization Intern May 2026 – Present
• Accelerated the convex QP behind IMRT radiotherapy planning, a path-following ADMM warm-starting an interior-point method on clinical instances up to **380K variables and 574K constraints**, replacing hand-tuning with **Bayesian optimization over 1,015 trials: 78 → 21 iterations to a usable warm start, 3.7× fewer and 3.2–3.8× less wall-clock** on a held-out clinical case
• Built the HPC platform behind **~1,100 solver runs and 1,015 parallel trials**: a matrix-free warm-started **conjugate-gradient GPU engine** (CuPy) with verified CPU/GPU numerical parity, **SLURM** job arrays over an NFS-safe journal, and a measurement-preserving early exit cutting trial cost **1.8–4.4×** while leaving reported results bit-identical
• **Derived and validated a predictive bound** explaining the gain: the capped penalty schedule, not the solver, was binding, imposing a floor of **69 of the prior method's 78 iterations**; confirmed against **ten one-knob ablations**

University of Southern California - Viterbi School of Engineering Los Angeles, CA
Graduate Student Researcher Aug 2025 – Present
• **Machine learning & NLP:** a 197K-parameter head steering **Mixture-of-Experts** routing at inference with **zero weight changes, +0.101 mean faithfulness** over six benchmarks (4× the ICLR 2026 baseline) and **+0.288** where the retrieved context contradicts what the model would otherwise say; sparse autoencoders (**93% variance explained**) over ~940K activations; first-author **AIIDE 2026** paper
• **Reinforcement learning and learning from demonstration:** trained agents with **distributed PPO/TRPO** under a **three-phase rising-opponent curriculum** across **100+ controlled runs**, and trained **behavior cloning and DAgger** policies in **MuJoCo**
• **Distributed systems:** a fault-tolerant **master-worker** service serving three model families across **14 GPU workers at ~32 req/s**, behind a **FastAPI** router with an **LRU cache**, per-worker concurrency caps for **backpressure**, and **health-checked service discovery**

Sharif University of Technology Tehran, TE
Research Assistant (Computer Vision) and Teaching Assistant Sep 2021 – Aug 2023
• Built **Vision Transformer and 3D CNN generative pipelines** on HPC/GPU clusters in a four-person computer-vision group (Dr. Tefagh); taught **300+ students** across 6 CS courses over 4 semesters

Projects

Search, Ranking and Recommendation Service (Personal) GitHub
• **128-dim user/item embeddings** over **3.1M ratings** (17,117 titles; power users with $\geq 1,200$ ratings, filtered from a **300K-user** corpus) serving **ranked top-N results** through a typed **REST API** with validated request schemas and a warm in-process cache
• Owned deployment end to end: a **7-stage Jenkins CI/CD pipeline** (build, push, rolling deploy) onto **Kubernetes**, with **DVC** versioning every dataset and model so any ranking run can be rebuilt or rolled back

Thread Pool and Resource-Aware Job Scheduler (Course) GitHub
• Built a **thread pool from scratch** in Java without `ExecutorService`, with runtime resizing, **resource-quota** admission, and a **lock-ordering discipline** that avoids deadlock, on a **1,000-task driver**

Multi-Modal Fusion for Sentiment Classification (Personal) GitHub
• Fused **RoBERTa, ViT, and wav2vec2** over text, image, and audio to reach **69% on MSCTD**, tracking every run on **SLURM** with **Weights & Biases** and serving the model behind **FastAPI**

Robot Localization from Noisy Sensor Data (Course) USC CSCI 545
• Wrote the sensor likelihood, motion model and importance resampling of a **2,000-particle Monte Carlo localizer** as a **ROS 1 node in Python**, validated on recorded **rosbag** runs

Publications

- **A Gold-Standard Study of What Makes a Lightweight Game-Playing Agent Strong.** First author. Accepted, **AIIDE 2026 (AAAI)**, Belo Horizonte, Nov 2026. arXiv:2607.06854
- **Inference-time steering of Mixture-of-Experts routing.** Sole author, manuscript in preparation (USC)

Leadership & Activities

- **Ranked 85th nationally (top 0.05%),** National University Entrance Exam (Math & Physics) | **11th of 1,200 teams,** Pardis Code programming contest | **Top 10 groups,** Mustafa Prize innovation competition
- **1st rank in department,** Sharif University | **Merit scholarship,** USC Viterbi | **3 merged pull requests to PettingZoo** (Farama Foundation) | **CITI Program** certification | **60+ public repositories**